

Lab 09

Multiplexer and Demultiplexer

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9.1 Objective

After performing this lab, students should be able to:

- Design and test Verilog modules for Multiplexer and Demultiplexer
- Identify the difference between Gate Level and Dataflow modeling
- Demonstrate data multiplexing through digital logic

9.2 Introduction

In this lab, you're going to build a system that is capable of driving 4 common anode seven segment displays with multiplexed data lines, to display different numbers on the each seven-segment LED display, we have to control the cathodes (YA-YG) of the four seven-segment LEDs separately and by activating the four seven-segment LEDs at different times. For example, when we activate Display 1 by driving the common anode terminal other three Displays (i.e., Display 2,3,4) will be deactivated using Enable lines, see Figure 9.1.

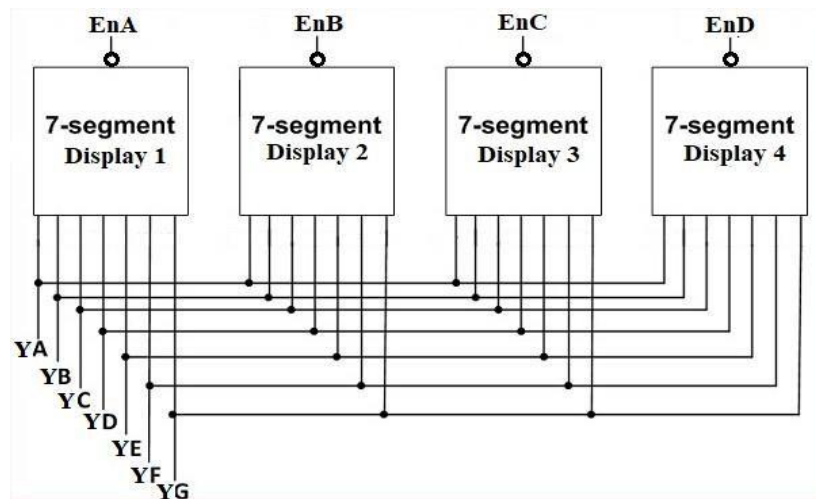


Figure 9.1

The overall system accepts 4-bit data in four data buses A, B, C and D for each seven-segment display, 2-bit data for the select line S to control the 4-bit data, which is passed to the binary to hexadecimal decoder1, the final output (7-bit data) of which goes to the seven-segment display and the seven segment LEDs are controlled by Enable bit to enable or disable the display of selected seven segment display.

Use Binary to hexadecimal module already created in Lab 6.

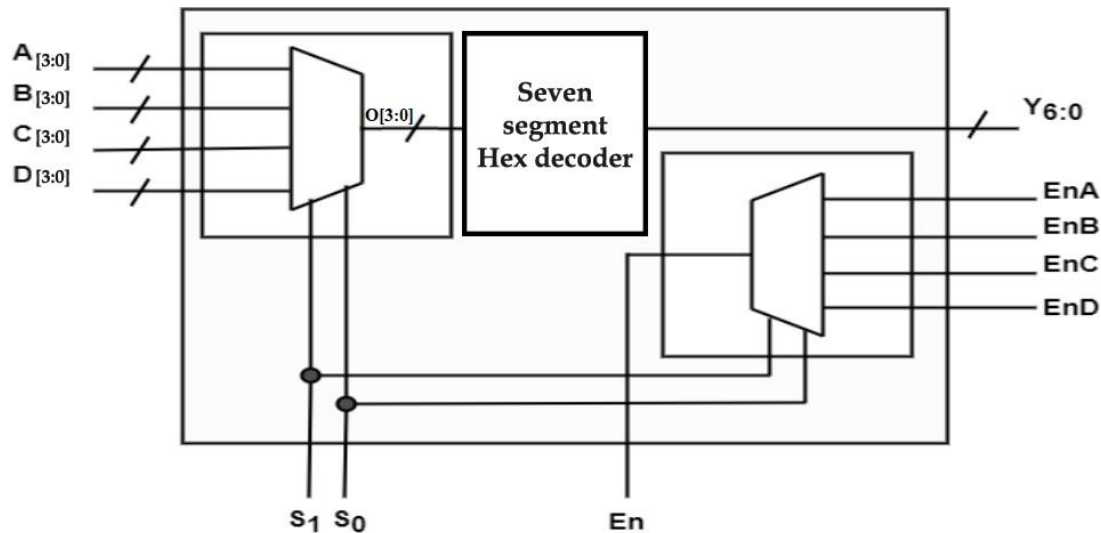


Figure 9.2 Top Level Module

In the next two sections, you'll implement the multiplexer and demultiplexer circuit and then connect them together to work as a single module.

9.3 Multiplexer

A multiplexer (MUX) is a combinational circuit that selects binary information from one of many input lines and directs it to a single output line. The selection of a particular input line is controlled by a set of selection lines. Normally, there are 2^n input lines and n selection lines whose bit combinations determine which input is selected and its logic state (i.e., 0 or 1) is transferred to output. Figure 9.3 shows the symbol of general multiplexer

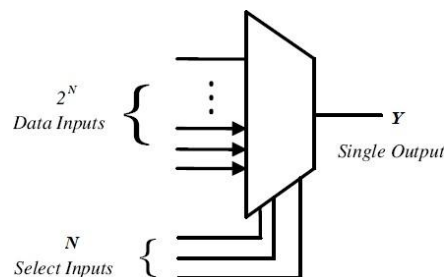


Figure 9.3

A two-to-one-line multiplexer connects one of two 1-bit sources to a common destination, as shown in Figure 9.4. The circuit has two data input lines, one output line, and one selection line S . When $S=0$, the upper AND gate is enabled and I_0 has a path to the output. When $S=1$, the lower AND gate is enabled and I_1 has a path to the output.

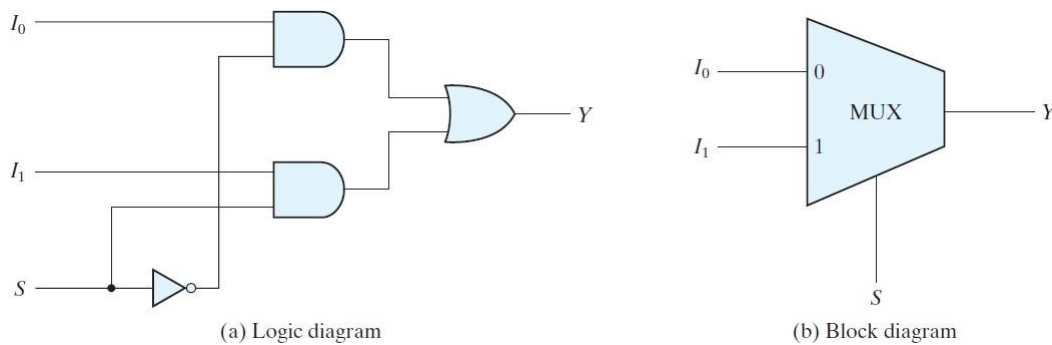


Figure 9.4

Implementations of 2:1 MUX

The simplest multiplexer is the **2:1 MUX** which simply selects its output from just two possible inputs. Its selection lines are therefore made of a single bit.

Table 9.1

S	I_0	I_1	Y
0	0	x	0
0	1	x	1
1	x	0	0
1	x	1	1

Note: 'x' in input columns as in Table 9.1 means, the variable can be 0 or 1. Do not confuse those as "don't care" conditions in k-maps which are represented as "X" in output column. The logic function for a 2:1 MUX can be derived using the min-terms from Table 9.1

$$(S, I_0, I_1) = (I_0 S) + (\overline{I_1} S)$$



Gate Level Vs Dataflow modeling

Gate Level Modeling is the low level or ‘Gate Level’ abstraction. At gate level, the circuit is described in terms of gates and their instantiation. Hardware designer in this level requires basic knowledge of digital logic design. Verilog supports basic logic gates as predefined “primitives”.

All logic circuits can be designed by using basic gates. For example

```
not M1(y, a);  
and M2(y, a, b);  
or M3(y, a, b);
```

Modeling in Gate level is very difficult for a complex circuit design. Data flow modeling provides a powerful way to implement a design. Verilog allows a circuit to be designed in terms of the dataflow between registers and process data or expression rather than instantiation of individual gates. This approach allows a designer to concentrate on optimizing the design in terms of data flow.

For maximum flexibility in the design process, designers typically use a Verilog description style that combine the concepts of Gate-level, and Data- flow design.

Conditional Operator

Conditional operators are frequently used in dataflow modeling to model conditional assignments. The conditional expression acts as a switching control.

The conditional operator (*? :*) takes three operands.

Usage: *condition_expr ? true_expr : false_expr;*

The condition expression (*condition_expr*) is first evaluated. If the result is true (logical 1), then the *true_expr* is evaluated. If the result is false (logical 0), then the *false_expr* is evaluated

If the result is *x* (ambiguous), then both *true_expr* and *false_expr* are evaluated and their results are compared, bit by bit, to return for each bit position an *x* if the bits are different and the value of the bits if they are the same.

The action of a conditional operator is similar to a multiplexer. Alternately, it can be compared to the if-else expression.

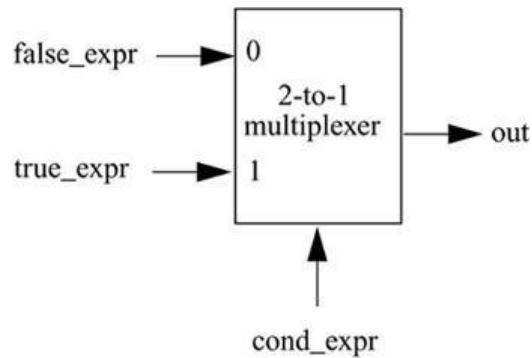


Figure 9.5

for example, 2:1 Mux in Figure 9.4 can be implemented as follows

```
//model functionality of a 2-to-1 mux assign
Y = S ? I1 : I0;
```

Task a)

In the previous lab (Lab 6) you have formulated a truth table for an active low decoder for (common anode) seven segment display, which will be used in the later part of this Lab. In this task you will develop a 4:1 MUX using conditional operator that will receive 3-bit binary data representing 4 digits of your registration ID and the output will be selected through the selection lines for each digit as shown in Figure 9.6.

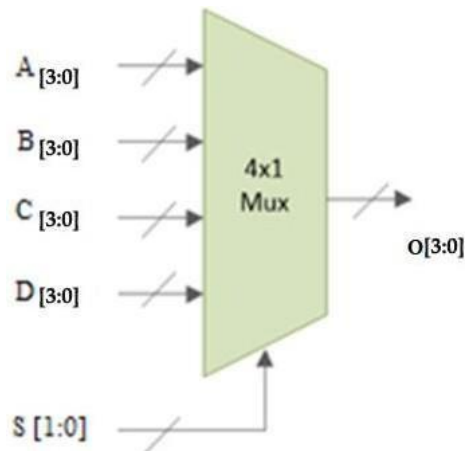


Figure 9.6

Conditional operations can be nested in two ways. Each true_expr or false_expr can itself be a conditional operation. For example



```
assign O = S[1] ? (S[0] ? D : C) : (S[0] ? B : A);
```

Or

```
assign O = (S==2'b00) ? A : (S==2'b01) ? B : (S==2'b10) ? C : D;
```

Complete the truth table provided in Table 9.2. As an example, when the Select line input is $S_1:0=00$, output will be the binary number stored in $A_3:0$

Table 9.2

S_1	S_0	$A_3:0$	$B_3:0$	$C_3:0$	$D_3:0$	$O_3:0$
0	0	1001	x	x	x	1001
0	1	x	0011	X	X	0011
1	0	x	x	0010	x	0010
1	1	X	X	X	0111	0111

Write a Verilog module for 4:1 MUX for Table 9.2 using conditional operator in Vivado and design a test bench to test the designed module for all possible values of S_1S_0 as per Table 9.2. Name your module as **mux_4_1**.

**CODE FOR MULTIPLEXER:-**

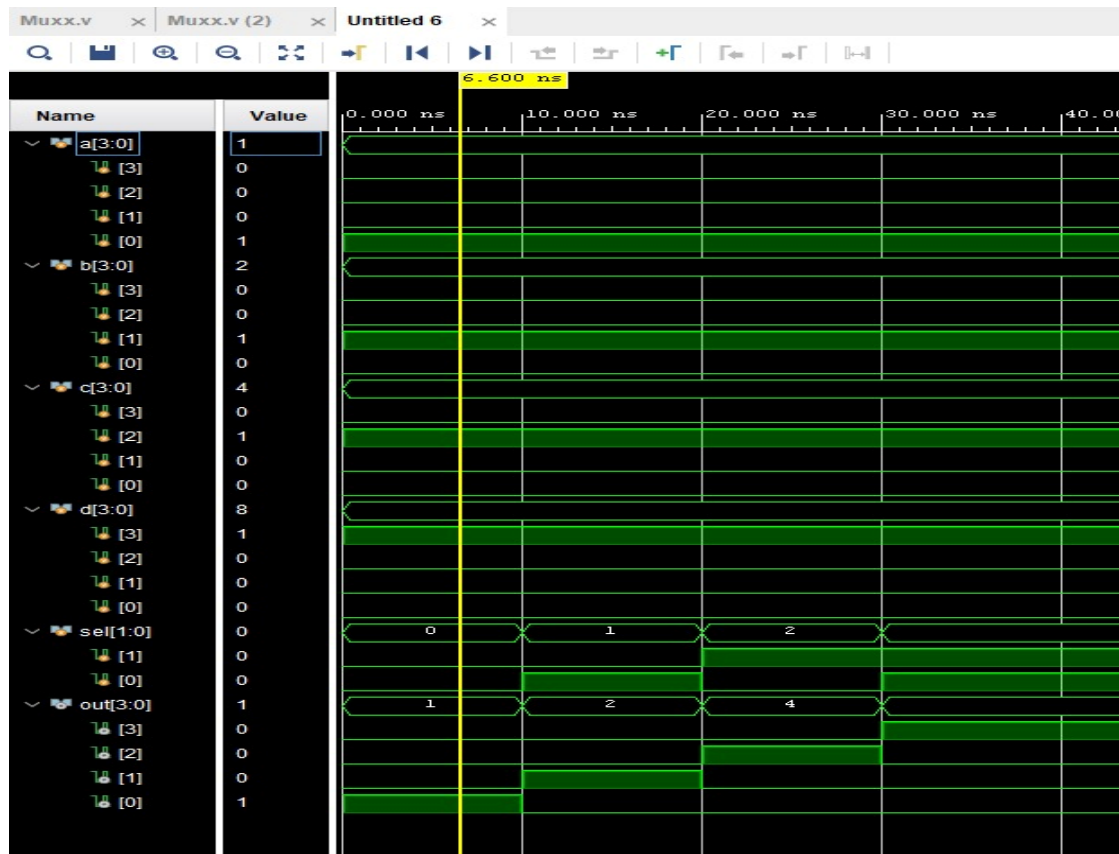
```
Muxx.v x Muxx.v (2) x Untitled 6 x
C:/Users/sa10303/Desktop/lab8/MUX/MUX.srcs/sources_1/new/Muxx.v
[Icons]
1  timescale 1ns / 1ps
2  //////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////
3  // Company:
4  // Engineer:
5  //
6  // Create Date: 03/07/2025 03:51:19 PM
7  // Design Name:
8  // Module Name: Muxx
9  // Project Name:
10 // Target Devices:
11 // Tool Versions:
12 // Description:
13 //
14 // Dependencies:
15 //
16 // Revision:
17 // Revision 0.01 - File Created
18 // Additional Comments:
19 //
20 //////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////
21
22
23 module Muxx(
24     input [3:0] a,
25     input [3:0] b,
26     input [3:0] c,
27     input [3:0] d,
28     input [1:0] sel,
29     output [3:0] out);
30
31     assign out = sel[1] ? (sel[0] ? d : c) : (sel[0] ? b : a);
32 endmodule
33
```

**CODE FOR TESTBENCH:**

```
Muxx.v x Muxx.v (2) x Untitled 6 x
C:/Users/sa10303/Desktop/lab8/MUX/MUX.srscs/sim_1/new/Muxx.v
[Icons]
21
22
23 module Muxx_testb();
24
25     reg [3:0] a;
26     reg [3:0] b;
27     reg [3:0] c;
28     reg [3:0] d;
29     reg [1:0] sel;
30     wire [3:0] out;
31
32     Muxx t1 (a, b, c, d, sel, out);
33
34     initial begin
35
36         a = 4'b0001;
37         b = 4'b0010;
38         c = 4'b0100;
39         d = 4'b1000;
40         sel = 2'b00;
41         #10;
42         sel = 2'b01;
43         #10;
44         sel = 2'b10;
45         #10;
46         sel = 2'b11;
47         #10;
48
49     end
50
51
52 endmodule
53
```

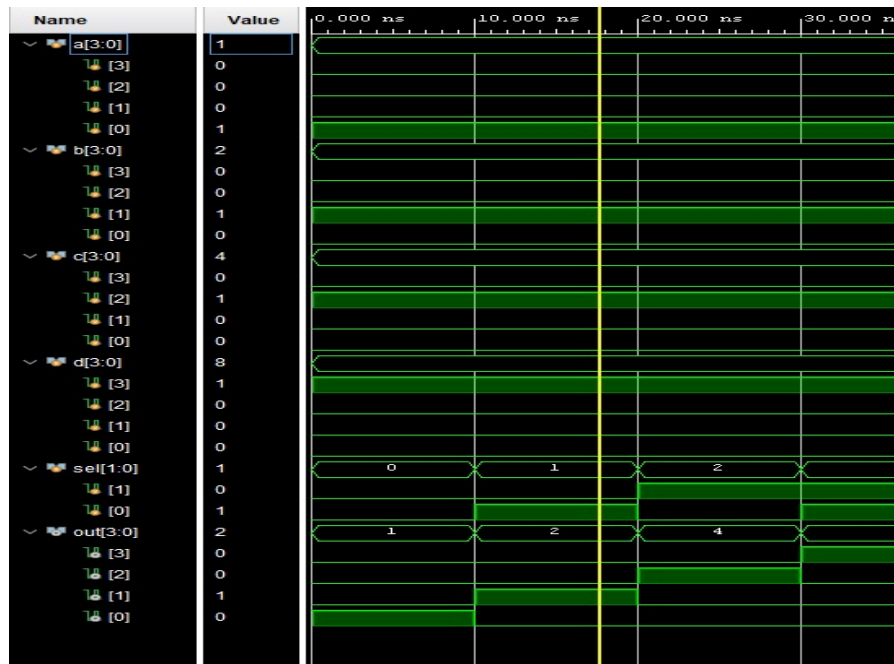


TEST 1: When the select is 00, the output is A, that is 0001:



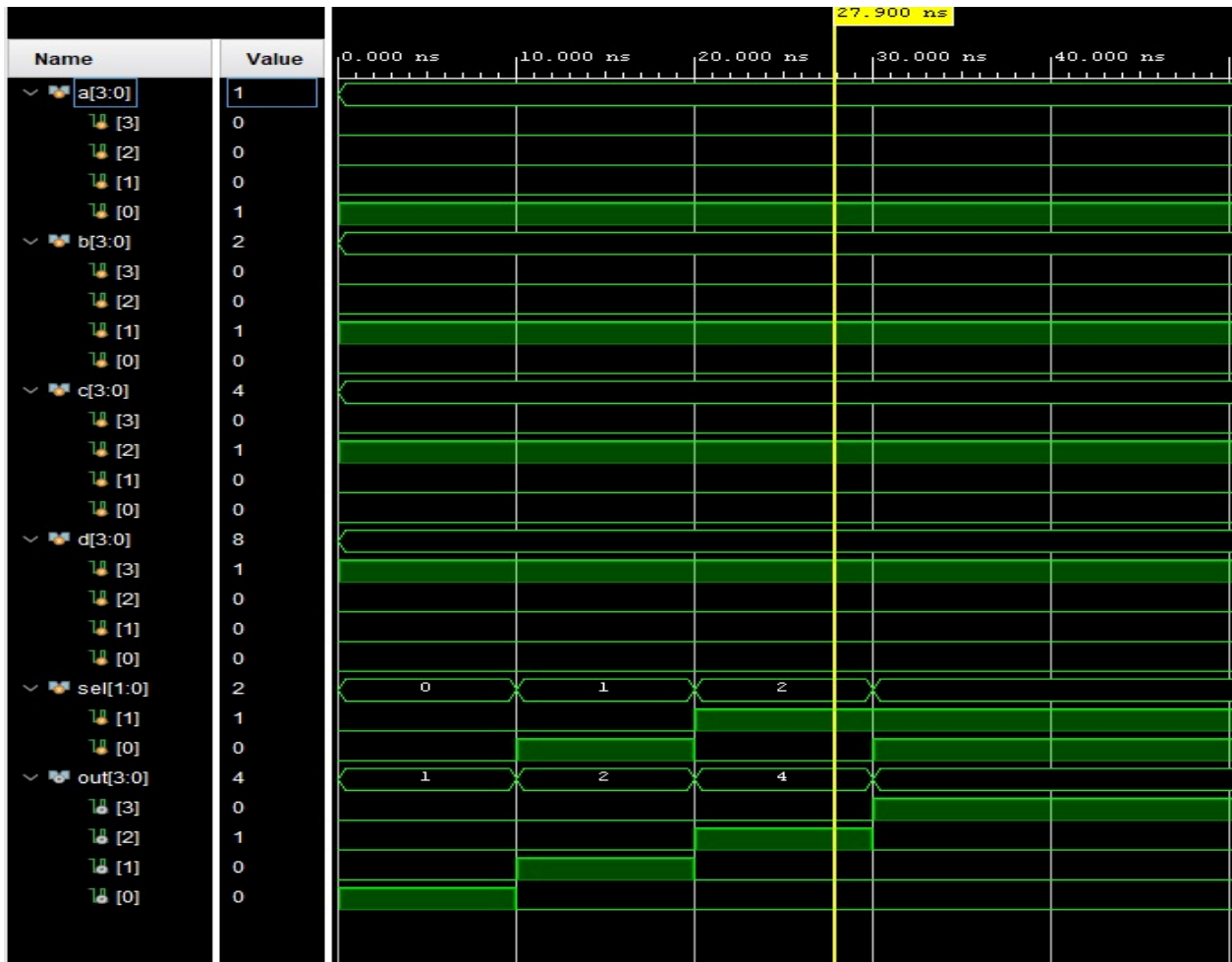


TEST 2: When the select is 01, Output is B that is 0010.

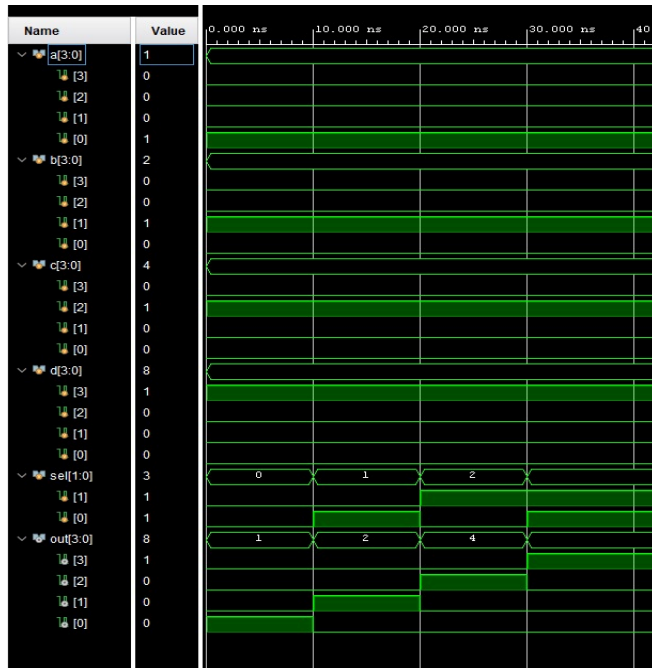




TEST 3: When the select is 10, output is c, that is 0100



TEST 4: WHEN THE SELECT IS 11, THE OUPUT IS D THAT IS 1000



9.4 Demultiplexer

The function of Demultiplexer (DeMUX) is opposite to multiplexer function. It takes information from one line and distributes it to a given number of output lines. For this reason, the Demultiplexer is also known as a data distributor.

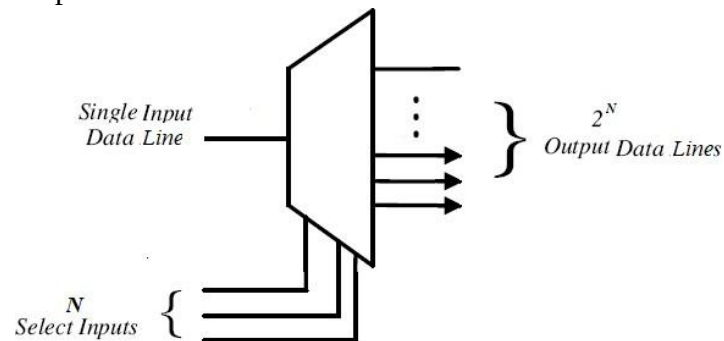


Figure 9.7

Implementations of 1:2 DeMUX

A 1:2 demultiplexer uses 1 select line (S) to determine which one of the 2 outputs (Y_0 – Y_1) is routed from the input (D). Its characteristics can be described in the following simplified truth table.

Table 9.3

S	D	Y_1	Y_0
0	0	0	0
0	1	0	1
1	0	0	0
1	1	1	0

The logic function for a 1:2 MUX can be derived using the min-terms from Table 9.3

$$Y_0 = S\bar{D}$$

$$Y_1 = SD$$

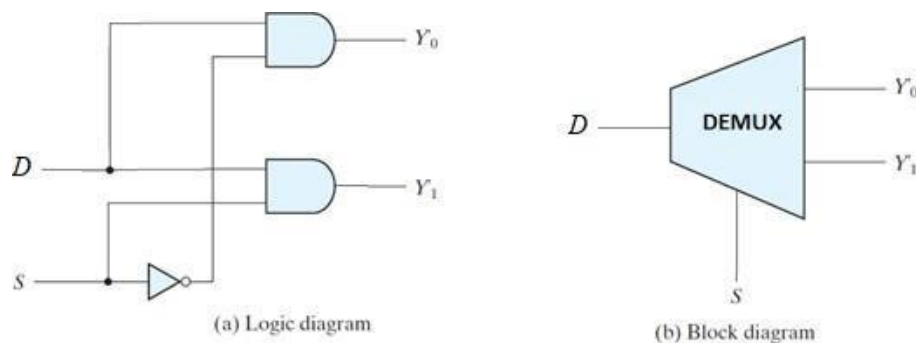
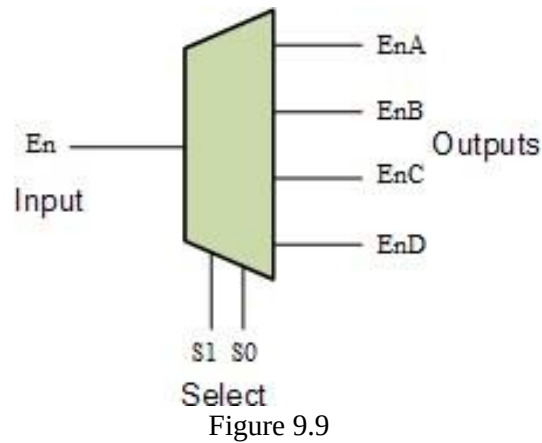


Figure 9.8

Task b)

So far, we have transmitted the data of our desired digit to the multiplexed data line $O[3:0]$ of binary to hexadecimal converter by selecting it through the select lines. In this task we will use 1:4 DeMux to enable one of 4 seven segment displays to show the result on seven-segment of our choice.

As discussed earlier each of the seven-segment display is Common Anode (CA), which require Logic 0 at the common terminal to work, but in figure 9.1 we can see an ‘inversion bubble’ at the CA terminal of each seven segment. Which mean we have to design a DeMux that work on active low logic.



Complete the truth table provided in Table 9.4. As an example, when the Select line input is $S_1:S_0=00$, output EnA will be 0 and the rest of the outputs will remain high.

Table 9.4

S_1	S_0	En	EnA	EnB	EnC	EnD
0	0	0	0	1	1	1
0	1	0	1	0	1	1
1	0	0	1	1	0	1
1	1	0	1	1	1	0
x	x	1	1	1	1	1



You can use conditional operator for each output equation such that assign `output_value1 = select_signal ? enable : 1'b1`

Write a Verilog module for 1:4 DeMUX for Table 9.4 using conditional operator in Xilinx Vivado and design a test bench to thoroughly test the designed module. Name your module as `demux_1_4`

**DEMUX MODULE DESIGN:-**

SIMULATION - Behavioral Simulation - Functional - sim_1 - Demux_testb

Demux.v x Demux_testb.v x Untitled 8 x

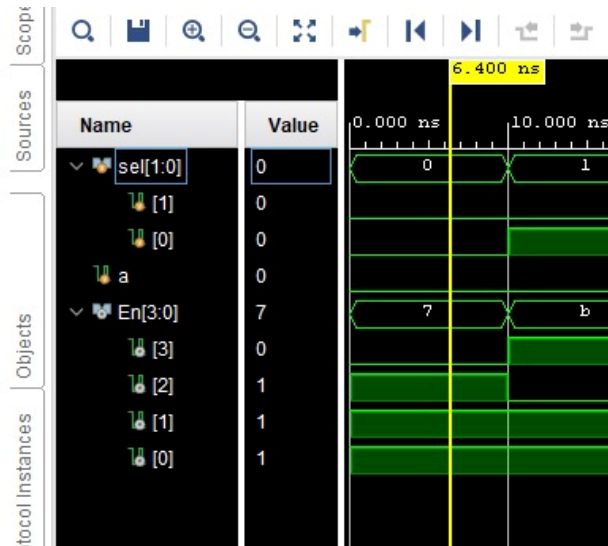
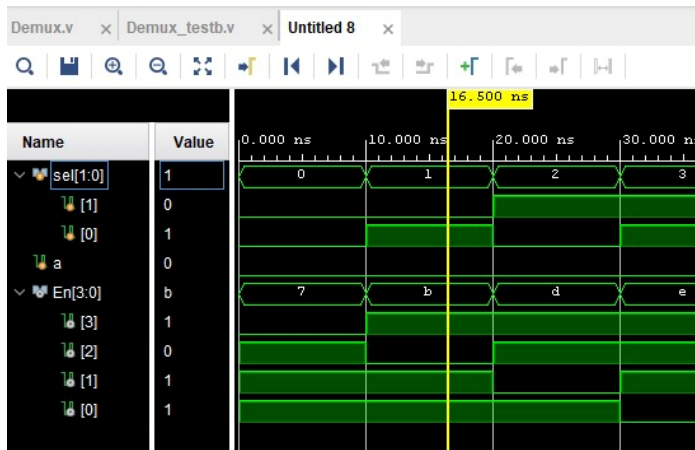
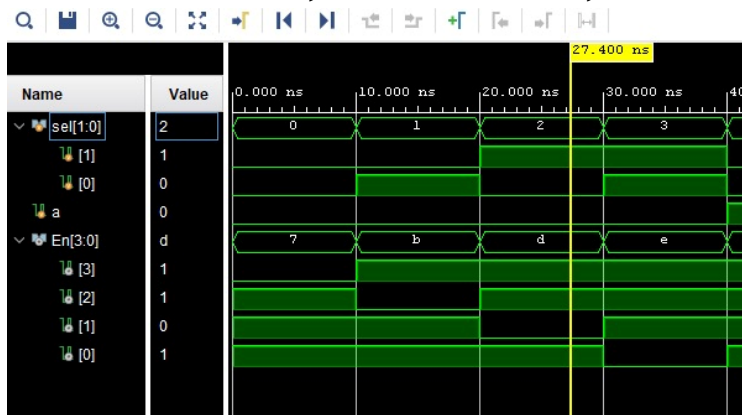
C:/Users/sa10303/Desktop/lab8/MUX/MUX.srscs/sources_1/new/Demux.v

1 timescale 1ns / 1ps
2
3 // Company:
4 // Engineer:
5 //
6 // Create Date: 03/07/2025 04:21:04 PM
7 // Design Name:
8 // Module Name: Demux
9 // Project Name:
10 // Target Devices:
11 // Tool Versions:
12 // Description:
13 //
14 // Dependencies:
15 //
16 // Revision:
17 // Revision 0.01 - File Created
18 // Additional Comments:
19 //
20
21
22
23 module Demux(
24 input [1:0] sel,
25 input a,
26 output [3:0] En);
27 assign En = (a==1'b0) ? (sel==2'b00) ? 4'b0111 : (sel==2'b01) ? 4'b1011 : (sel==2'b10) ? 4'b1101 : 4'b1110 : 4'b1111;
28
29 endmodule
30

**DEMUX TEST BENCH:-**

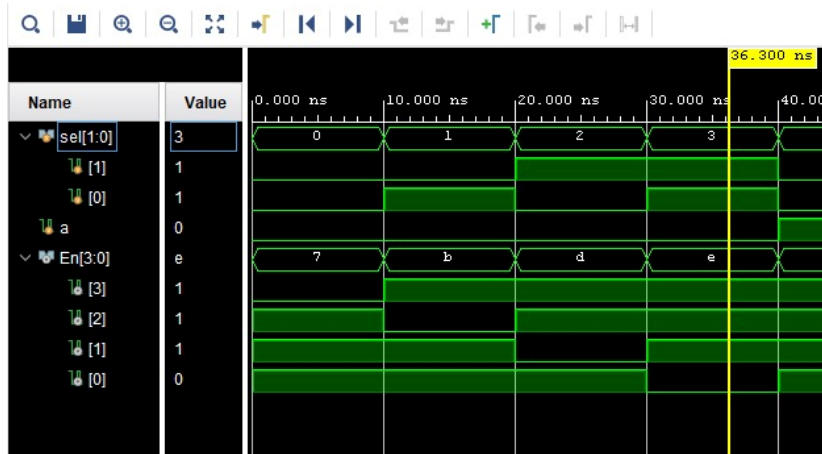
```
Demux.v x Demux_testb.v x Untitled 8 x
C:/Users/sa10303/Desktop/lab8/MUX/MUX.srscs/sim_1/new/Demux_testb.v

11 // Tool Versions:
12 // Description:
13 //
14 // Dependencies:
15 //
16 // Revision:
17 // Revision 0.01 - File Created
18 // Additional Comments:
19 //
20 //////////////////////////////////////
21
22 module Demux_testb();
23
24     reg [1:0] sel;
25     reg a;
26     wire [3:0] En;
27
28     Demux t1 (sel, a, En);
29
30     initial begin
31
32         a = 1'b0;
33         sel = 2'b00;
34         #10;
35         sel = 2'b01;
36         #10;
37         sel = 2'b10;
38         #10;
39         sel = 2'b11;
40         #10;
41
42         a = 1'b1;
43         sel = 2'b00;
44         #10;
45         sel = 2'b01;
46         #10;
47         sel = 2'b10;
48         #10;
49         sel = 2'b11;
50         #10;
51
52     end
53 endmodule
```

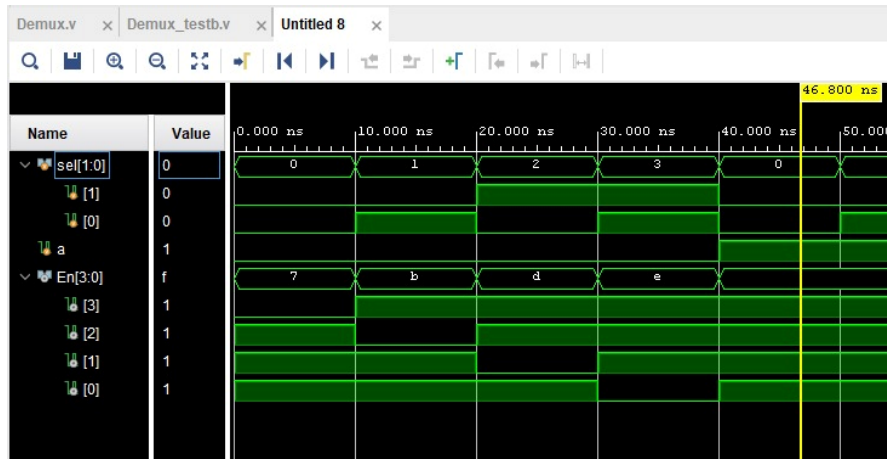
**TEST 1:- WHEN A=0, AND SELECT =00, OUTPUT IS 0111****TEST 2:- WHEN A=0, AND SELECT =01, OUTPUT IS 1011****TEST 3:- WHEN A=0, AND SELECT =10, OUTPUT IS 1101**



TEST 4:- WHEN A=0, AND SELECT=11, OUTPUT IS 1110



TEST 5: WHEN ENABLE=1: ALL THE INPUTS ARE 1111



**Task c)**

Write a Verilog module `topLevelModule` that combines MUX (**`mux_4_1`**) and DeMUX (**`demux_1_4`**) modules created in this lab and binary-hexadecimal decoder module already created in Lab 6, according to the block diagram of Figure 9.2 Top Level Module. This module's inputs includes four 4-bit data lines (i.e., `A[3:0]`, `B[3:0]`, `C[3:0]`, `D[3:0]`), 2-bit select line (i.e. `S[1:0]`) to select from the data for binary-hexadecimal decoder, and a 1-bit active low enable signal (i.e. `En`). The modules output includes one 7-bit data line for seven segments (i.e., `Y [6:0]`) and four 1-bit Enable signals (i.e., `EnA`, `EnB`, `EnC`, `EnD`).

9.5 Exercise (Post Lab)**a) Add code of the `topLevelModule`**

```
input En,
output [6:0] O,
output anode,
output [2:0] High);

wire EnA;
wire EnB;
wire EnC;
wire EnD;

wire [3:0] O1;

reg [3:0] D = 4'b0100;

assign High = 3'b011;

Mux m(A,B,C,D,S,O1);

segmentDecoder s(O1,O);

Demux d(En,S,EnA,EnB,EnC,EnD);

assign anode = (EnA==0)?EnA:
               (EnB==0)?EnB:
               (EnC==0)?EnC:
               (EnD == 0)? EnD: 1;

endmodule
```



- b) Create testbench for your topLevelModule and test each line by holding binary coded decimal number of your student registration ID.

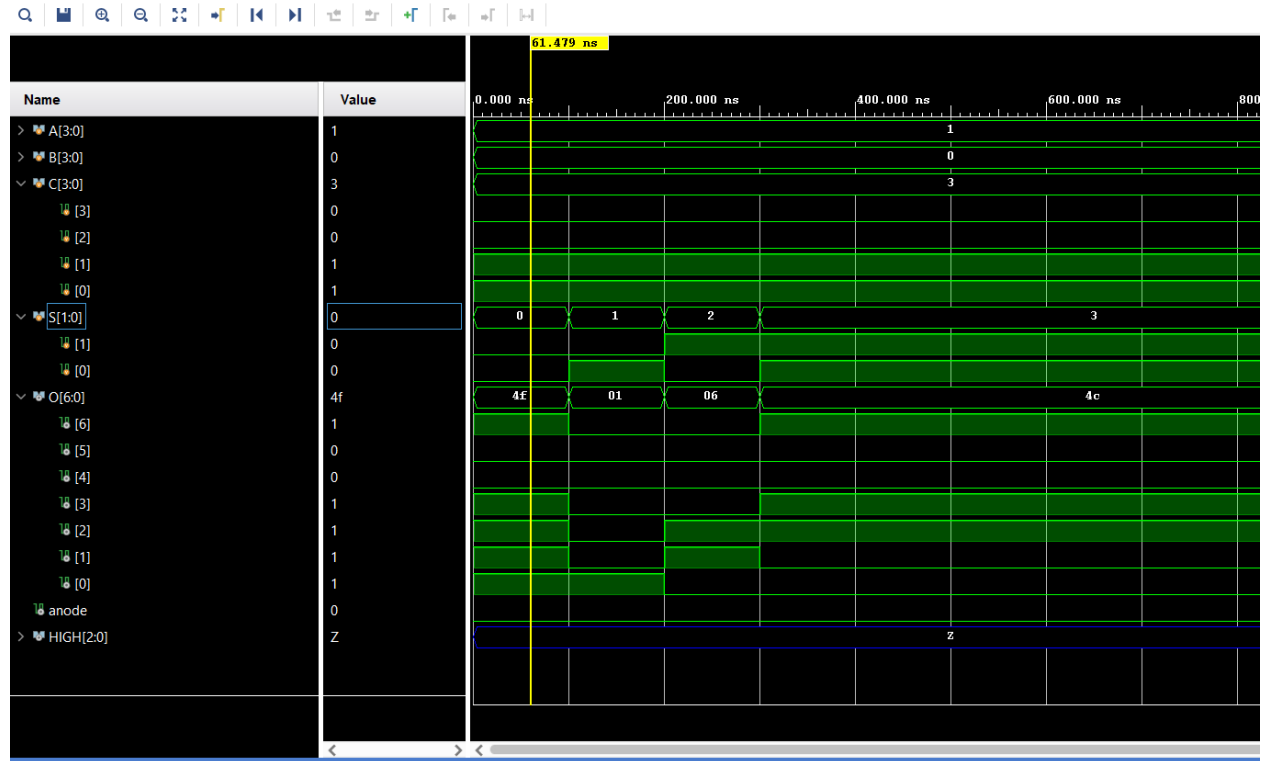
```
reg [3:0] B = 4'b0000;
reg [3:0] C = 4'b0011;
reg [1:0] S;
wire [6:0] O;
wire anode;
wire [2:0] HIGH;
TopLevelModule test_toplevelmodule(A,B,C,S,0,O,anode);

)    initial
)    begin
    S = 2'b00;
    #100
    S = 2'b01;
    #100
    S = 2'b10;
    #100
    S = 2'b11;
)    end
) endmodule
```

c)



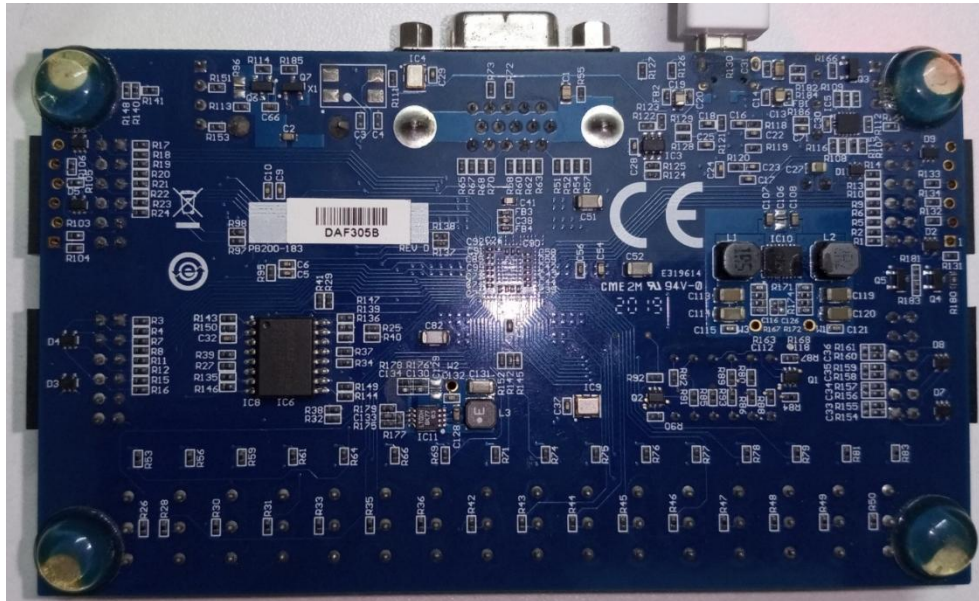
Student id is 10303: a=1, b=0, c=3 and the last one was hard coded as we don't have enough inputs on the fpga.



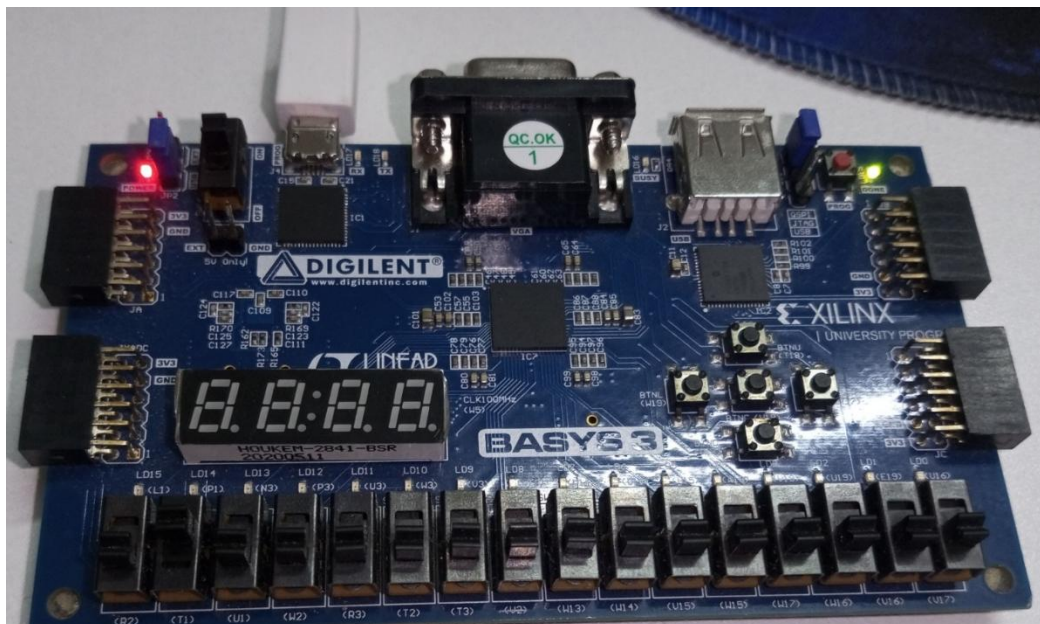
d)

- e) Implement your design on BASYS-3 board to demonstrate data multiplexing. Note that, on Basys-3 board there are 16 input switches and the total number of input bits in this design are nineteen (19), in order to test your code on board hardcode input D[3:0] in your Verilog topLevelModule.

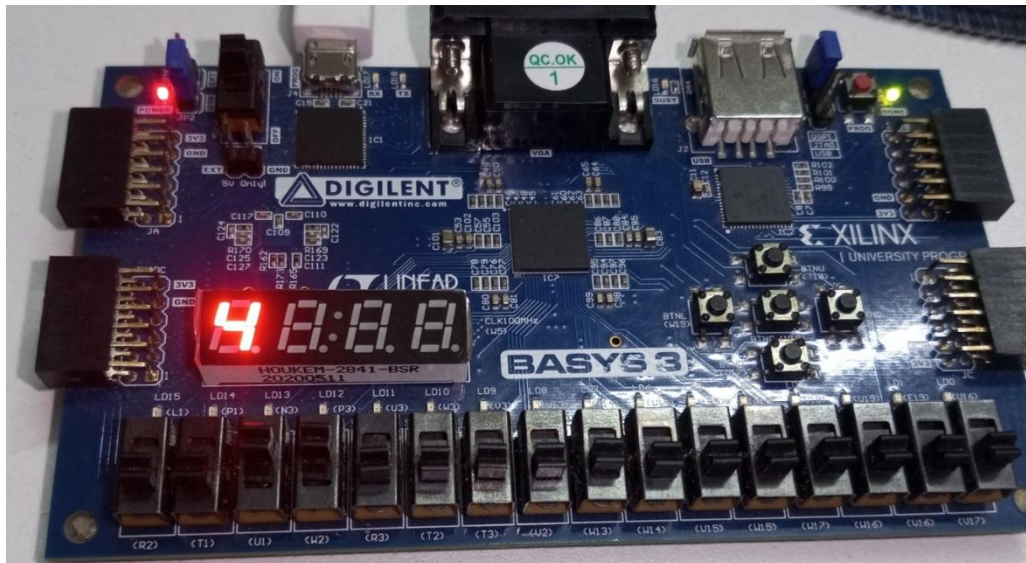
HARD CODED THE VALUE OF D=4. (board number:- DAF305B)



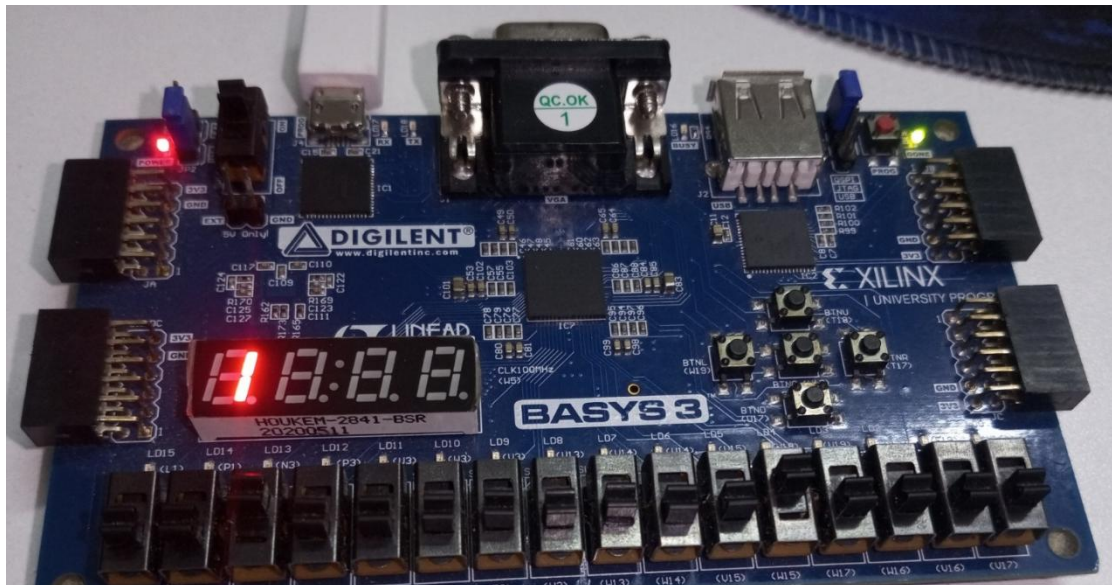
WHEN ENABLE IS ZERO THERE IS NO OUTPUT:-



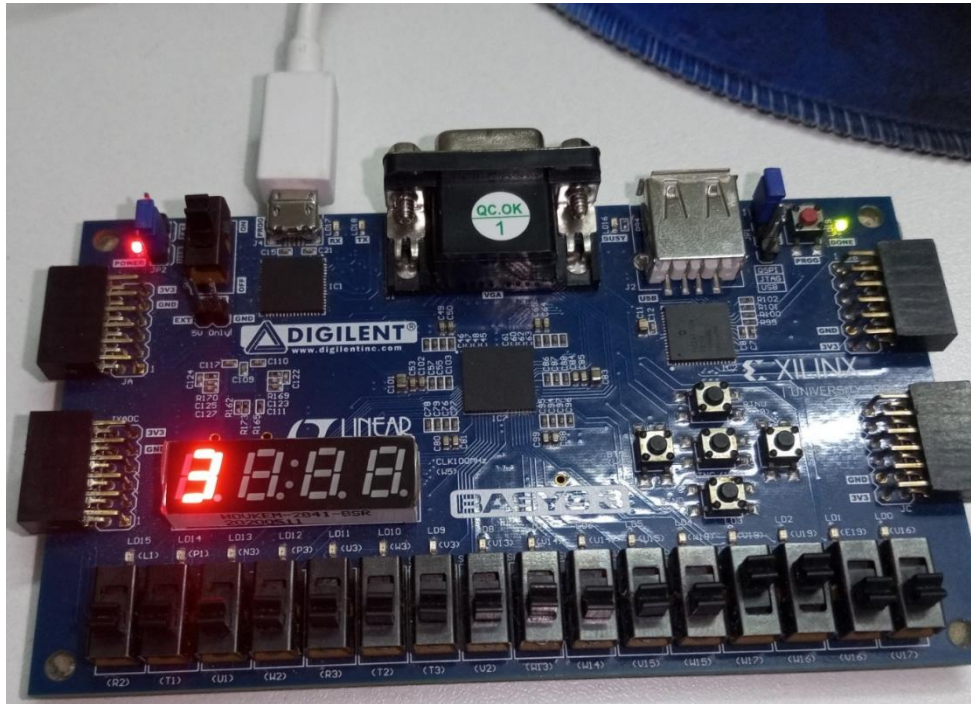
FOR $S_0=1$ AND $S_1=1$, OUTPUT MUST BE D, WHICH IS 4



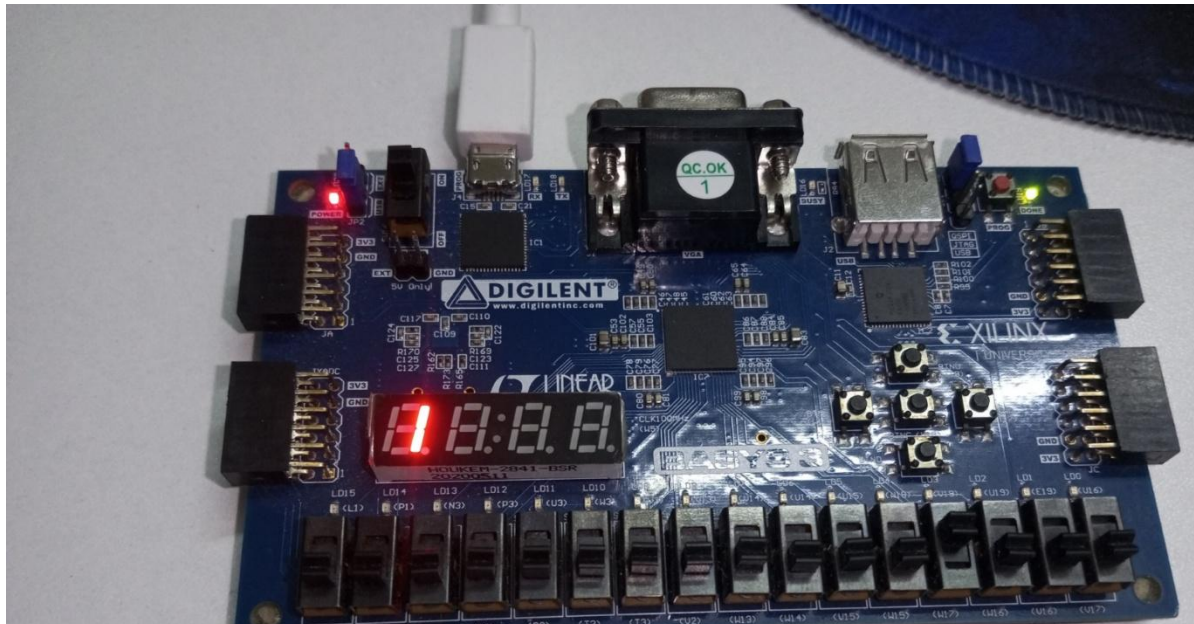
FOR $S_0=1$ AND $S_1=0$, THE OUTPUT MUST BE B, WHICH IS 0001:



FOR $S_0=0$ AND $S_1=0$, THE OUTPUT MUST BE A, WHICH IS 0011:



HERE $A=0001$





Assessment Rubric
Lab 09
Multiplexer and Demultiplexer

Name:	Student ID:
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Marks Distribution:

		LR2 Code	LR5 Results	LR9 Report
In - Lab	Task a	10	10	20
	Task b	10	10	
	Task c	10	5	
Exercise		10	5	
Total Marks: 100		/40	/30	/20
CLO Mapping		CLO2	CLO2	CLO4

Affective Domain Rubric		Points	CLO Mapped
AR 7	Report Submission and Viva	/10	CLO 4

CLO	Total Points	Points Obtained
2 (Psy – 3)	70	
4 (Aff-3)	30	



Total	100	
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For description of different levels of the mapped rubrics, please refer to the provided Lab Evaluation Assessment Rubrics and Affective Domain Assessment Rubrics.

Lab Evaluation Rubrics

#	Assessment Elements	Level 1: Unsatisfactory	Level 2: Developing	Level 3: Good	Level 4: Exemplary
LR2	Program/Code/ Simulation Model/ Network Model	Program/code/simulation model/network model does not implement the required functionality and has several errors. The student is not able to utilize even the basic tools of the software.	Program/code/simulation model/network model has some errors and does not produce completely accurate results. Student has limited command on the basic tools of the software.	Program/code/simulation model/network model gives correct output but not efficiently implemented or implemented by computationally complex routine.	Program/code/simulation /network model is efficiently implemented and gives correct output. Student has full command on the basic tools of the software.
LR5	Results & Plots	Figures/ graphs / tables are not developed or are poorly constructed with erroneous results. Titles, captions, units are not mentioned. Data is presented in an obscure manner.	Figures, graphs and tables are drawn but contain errors. Titles, captions, units are not accurate. Data presentation is not too clear.	All figures, graphs, tables are correctly drawn but contain minor errors or some of the details are missing.	Figures / graphs / tables are correctly drawn and appropriate titles/captions and proper units are mentioned. Data presentation is systematic.
LR7	Viva	Response shows a complete lack of understanding of the assigned / completed task	Response shows shallow understanding of the assigned task	Response shows substantial understanding of the assigned task	Response shows complete understanding of the completed task. The student is able to explain all the related concepts.



LR9	Report	No summary provided. The number/amount of tasks completed below the level of satisfaction and/or submitted late	Couldn't provide good summary of in-lab tasks. Some major tasks were completed but not explained well. Submission on time. Some major plots and figures provided	Good summary of In-lab tasks. All major tasks completed except few minor ones. The work is supported by some decent explanations, Submission on time, All necessary plots, and figures provided	Outstanding Summary of In-Lab tasks. All task completed and explained well, submitted on time, good presentation of plots and figure with proper label, titles and captions
-----	--------	---	--	---	---